

Lab 01: The Rental Refund Crisis

Exercise 1: Testing the "All or Nothing"

Query

Query History

```
1 SELECT * FROM rental
```

Data Output

Messages

Notifications

≡+

▼

▼

SQL

Showing rows: 1 to 1000

Page No: 1

	rental_id [PK] integer	rental_date timestamp without time zone	inventory_id integer	customer_id smallint	return_date timestamp without time zone	staff_id smallint	last_update timestamp without time zone
1	2	2005-05-24 22:54:33	1525	459	2005-05-28 19:40:33	1	2006-02-16 02:30:53
2	3	2005-05-24 23:03:39	1711	408	2005-06-01 22:12:39	1	2006-02-16 02:30:53

```
Query History
1 BEGIN;
2 UPDATE rental
3 SET return_date = NOW()
4 WHERE rental_id = 2;
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
UPDATE 1

Query returned successfully in 46 msec.

```
5 SELECT return_date
6 FROM rental
7 WHERE rental_id = 2;
```

Data Output Messages Notifications

Showing rows: 1 to 1

	return_date timestamp without time zone
1	2026-09-18 14:48:25.511765

QueryQuery History

```
1 BEGIN;
2
3 -- Step 1: Mark as returned
4 UPDATE rental
5 SET return_date = NOW()
6 WHERE rental_id = 2;
7
8 -- Step 2: Intentional Error
9 INSERT INTO payment (
10     customer_id,
11     staff_id,
12     rental_id,
13     amount,
14     payment_date
15 )
16 VALUES (
17     459,
18     1,
19     2,
20     99.99,
21     'WRONG_DATE_FORMAT'
22 );
23
24 COMMIT;
```

Data OutputMessagesNotifications

ERROR: invalid input syntax for type timestamp: "WRONG_DATE_FORMAT"
LINE 21: 'WRONG_DATE_FORMAT'
^

```
1 SELECT return_date
2 FROM rental
3 WHERE rental_id = 2;
```

Data OutputMessagesNotifications

Showing rows: 1 to 1

Page No: 1

	return_date timestamp without time zone
1	2005-05-28 19:40:33

```

2 FROM payment
3 WHERE rental_id = 2
4 AND amount = 99.99;

```

Data Output Messages Notifications

	payment_id [PK] integer	customer_id smallint	staff_id smallint	rental_id integer	amount numeric (5,2)	payme timest
--	----------------------------	-------------------------	----------------------	----------------------	-------------------------	-----------------

No, return_date is not updated because the invalid payment_date causes the INSERT to fail. PostgreSQL rolls back the entire transaction, including the previous UPDATE, to maintain Atomicity.

Savepoints: Partial Rollbacks

```

1 BEGIN;
2
3 INSERT INTO customer (
4     first_name,
5     last_name,
6     email,
7     address_id,
8     store_id
9 )
10 VALUES (
11     'AI',
12     'Student',
13     'ai@edu.vn',
14     1,
15     1
16 );

```

Data Output Messages Notifications

INSERT 0 1

Query returned successfully in 67 msec.

Query Query History

```
1 SAVEPOINT user_created;
```

Data Output Messages Notifications

SAVEPOINT

Query returned successfully in 43 msec.

Query Query History

```
1 INSERT INTO special_offers (customer_id, code)
2 VALUES (9999, 'FREE');
```

Data Output Messages Notifications

ERROR: relation "special_offers" does not exist
LINE 1: INSERT INTO special_offers (customer_id, code)
 ^

SQL state: 42P01

Character: 13

```
1 ROLLBACK TO SAVEPOINT user_created;
```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 49 msec.

Query Query History

```
1 SELECT
2     customer_id,
3     first_name,
4     last_name,
5     email
6 FROM customer
7 WHERE email = 'ai@edu.vn';
```

Data Output Messages Notifications

	customer_id [PK] integer	first_name character varying (45)	last_name character varying (45)	email character varying (50)
1	600	AI	Student	ai@edu.vn

Exercise 2: The "Flexible" Staff Task

Query

Query History

1

SELECT store_id

2

FROM store

3

ORDER BY store_id;

4

5

SELECT address_id

6

FROM address

7

ORDER BY address_id

8

LIMIT 5;

Data Output

Messages

Notification

Showing rows: 1 to 5

Page No:

	address_id [PK] integer
1	1
2	2
3	3
4	4
5	5

QueryQuery History

```
1 BEGIN;
2
3 INSERT INTO staff (
4     first_name,
5     last_name,
6     address_id,
7     email,
8     store_id,
9     active,
10    username
11 )
12 VALUES (
13     'Lab',
14     'Student',
15     1,
16     'lab.staff01@example.com',
17     1,
18     TRUE,
19     'lab_staff_01'
20 );
21
22 SAVEPOINT staff_created;
```

Data OutputMessagesNotifications

SAVEPOINT

Query returned successfully in 119 msec.

QueryQuery History

```
1 SELECT
2     staff_id,
3     first_name,
4     last_name,
5     store_id,
6     username
7 FROM staff
8 WHERE username = 'lab_staff_01';
```

Data OutputMessagesNotifications

SQL

Showing rows: 1 to 1

	staff_id [PK] integer	first_name character varying (45)	last_name character varying (45)	store_id smallint	username character varying (16)
1	3	Lab	Student	1	lab_staff_01

Query Query History

```
1 UPDATE staff
2 SET store_id = 'NOT_A_NUMBER'
3 WHERE username = 'lab_staff_01';
```

Data Output Messages Notifications

ERROR: invalid input syntax for type smallint: "NOT_A_NUMBER"
LINE 2: SET store_id = 'NOT_A_NUMBER'
^

SQL state: 22P02

Character: 29

Query Query History

```
1 ROLLBACK TO SAVEPOINT staff_created;
```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 67 msec.

Query Query History

```
1 UPDATE staff
2 SET store_id = 1
3 WHERE username = 'lab_staff_01';
4
5 COMMIT;
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 47 msec.

Query Query History

```
1 SELECT
2   staff_id,
3   first_name,
4   last_name,
5   store_id,
6   username
7 FROM staff
8 WHERE username = 'lab_staff_01';
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1 rows					
	staff_id [PK] integer	first_name character varying (45)	last_name character varying (45)	store_id smallint	username character varying (16)
1	3	Lab	Student	1	lab_staff_01